



Public Perception of Cultural Ecosystem Services in the Conservation Forest Area of Weh Island, Aceh

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Abstract: Pulau Weh, with Sabang as its capital, is located at the westernmost tip of Indonesia. Beyond its significance for domestic and international tourism, the island hosts several national strategic projects. Currently, the landscape of Pulau Weh is dominated by non-urban areas (agriculture, forests, and open land), including critical conservation and protected forest zones. These areas provide essential Cultural Ecosystem Services (CES) that benefit the local island community. This study aims to analyze land-use changes and local community perceptions of CES values within the forest conservation areas of Pulau Weh, Aceh Province. The research aligns with the Sustainable Development Goals (SDGs), specifically Goals 1 (No Poverty), 11 (Sustainable Cities and Communities), and 13 (Climate Action). A mixed-methods approach was employed, integrating ArcGIS analysis to evaluate spatial functional changes and statistical analysis using Principal Component Analysis (PCA) via SPSS to measure public perception. The results indicate that the 2023 land-use composition comprises built-up areas (497.69 ha; 4.49%), woodlands (5,203 ha; 46.89%), agricultural land (3,111 ha; 28.04%), grasslands (2,071 ha; 18.67%), water bodies (157 ha; 1.41%), and bare land (55.76 ha; 0.50%). Based on community perceptions, the forest conservation areas in Sabang contribute significantly to CES values, including recreation, cultural heritage, aesthetics, education, social relations, health, spirituality, and ecological/tourism benefits. Notably, disaster mitigation was the only value perceived as non-significant by the community.

Keywords: Weh Island; Cultural Ecosystem Services; Community perception; Conservation forest; Land use and land cover

1 Introduction

Pulau Weh, with Sabang as its capital, is the westernmost territory of Indonesia and an island city located in the province of Aceh [1]. As an archipelagic city, its urban footprint is significantly smaller than its non-urban areas, comprising agriculture, forests, and open land, positioning it as a potential garden city that attracts both domestic and international tourists. The island also encompasses substantial conservation and protected forest zones [2]. According to Law No. 41/1999 on Forestry, conservation forests are defined as forest areas with specific characteristics primarily functioning to preserve plant and animal diversity and their respective ecosystems. Furthermore, the Regional Spatial Plan (Qanun No. 6/2012) for Sabang (2012–2032) designates protected forests in the Aneuk Laot Lake and Kilometer Zero Monument areas, while conservation forests are distributed across the island, including Anoe Itam, Jaboi, Iboih, and Gapang [3]. These conservation forests are scattered throughout the island, often situated in coastal zones, adjacent to residential areas, and within emerging "new villages," resulting in highly dynamic environmental conditions.

As a small island characterized by complex ecosystem dynamics and community life, Pulau Weh is experiencing shifts in forest management from a regional landscape perspective, particularly since its designation as a National Strategic Activity Center (NSAC) within the national urban system [4]. The locations are illustrated in Figure 1. In the context of small island studies, sufficient attention to land use and land cover (LULC) changes and their subsequent impacts on ecosystems and local communities is often lacking [5]. For a sustainable urban system, a

balanced LULC planning approach is imperative to ensure that the benefits for the community and biodiversity can be maintained in the long term [6].



Figure 1. Study area map

Source: https://www.nationsonline.org/oneworld/map/indonesia_map.htm

The conservation forests distributed across Pulau Weh are increasingly utilized for built-up development, albeit with an emphasis on environmental sustainability. The designation of Sabang as an NSAC has intensified the demand for infrastructure [3], necessitating a meticulous balance in environmental management, specifically regarding land cover. For instance, mangrove rehabilitation in the Jaboi area is conducted annually to mitigate coastal abrasion; however, the success rate of these rehabilitation efforts remains low. Currently, the Sabang municipal government and relevant stakeholders continue to promote reforestation in the Jaboi coastal area and other regions. The locations of the conservation forests are illustrated in Figure 2. Specifically, the conservation forests in Pulau Weh can be categorized into three typologies:

- Conservation forests with significant land-cover growth, such as Sumur Tiga and Anoe Hitam;
 - Conservation forests proximal to residential settlements, including Cot Baku, Ujong Kareung, and Balee Pasie;
- and
- Conservation forests impacted by urbanization, such as Jaboi, Iboih, and Gapang.

The function of forests as environmental buffers is critical, as they are expected to provide substantial ecosystem service values essential to human well-being [7]. Cultural Ecosystem Services (CES) represent the non-material benefits that people obtain from ecosystems through spiritual enrichment, cognitive development, reflection, recreation, and aesthetic experiences [8]. In the context of developing nations, communities often place a higher premium on CES because cultural identity is deeply intertwined with their livelihoods and socio-ecological resilience in the Millenium Ecosystem Assessment (MEA, 2005) framework [9, 10]. Furthermore, the integration of green and blue spaces constitutes a vital dimension in assessing the holistic value of CES within a landscape [11].

These non-material ecosystem services represent a vital metric for predicting the environmental sustainability of future urban landscapes [12, 13]. Consequently, an in-depth assessment of CES within conservation forest areas is imperative to safeguard the existence and long-term viability of these ecosystems. CES enriches spiritual values, recreation, cognitive development, reflection, and aesthetic experiences, serving as a fundamental non-material

conduit between human societies and their surrounding ecosystems [14, 15]. Furthermore, the CES values and their operational details are presented in Table 1 below.

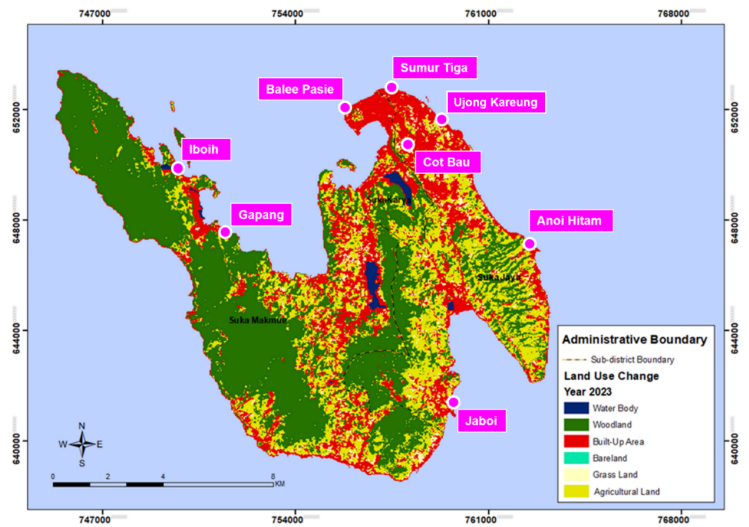


Figure 2. Location and spatial distribution of land use and land cover (LULC) in Weh/Sabang City, 2023

Table 1. CES values and their operational definition

No.	Components/Variables	Operational Definitions
1	Recreational value	Facilitates outdoor activities for leisure and relaxation, such as walking and general open-air recreation.
2	Cultural heritage value	Preserves cultural legacy, including historical remains, traditional practices, and cultural services.
3	Aesthetic value	Facilitates the enjoyment of ecosystem aesthetics, encompassing visual satisfaction, artistic inspiration, and the psychological stimulation derived from natural landscapes.
4	Educational value	Provides opportunities for scientific knowledge production, research, experiential learning, and formal/informal education through the natural environment.
5	Social relations value	Offers communal spaces that foster social cohesion, community building, and local cooperation.
6	Health value	Provides spaces for routine physical and mental activities that enhance overall physiological and psychological well-being.
7	Spiritual value	Provides spiritual stability and sanctuary for self-reflection and the cultivation of gratitude for divine creation.
8	DMV	Facilitates community safety by providing accessible spaces for self-rescue and designated evacuation sites.
9	Ecological/environmental value	Provides a comfortable and shaded microclimate, acting as “urban lungs” for the absorption of pollution and serving as a vital habitat for flora and fauna.
10	Tourism value	Offers a destination for nature-based tourism that promotes psychological well-being through scenic landscapes and the appreciation of biodiversity.

Note: CES = Cultural Ecosystem Services; DMV = Disaster mitigation value.

The community of Pulau Weh possesses a limited technical understanding of the term CES; however, they exhibit a profound practical awareness of the direct benefits provided by forest conservation areas. These forests provide essential spaces for community activities and offer tangible environmental benefits that are felt by the local

inhabitants [2]. CES is fundamental to the sustainability of urban populations [8]. For instance, research conducted in Gwacheon, Republic of Korea, demonstrated the utility of CES through a framework of seven cultural service categories within urban green spaces: recreation, cultural heritage, aesthetic value, educational value, spiritual value, health value, and social relations [16, 17].

In this study, the CES components for Pulau Weh/Sabang were expanded into ten distinct benefit categories, incorporating three additional values beyond the standard MEA (2005) framework [18]. These three additions, disaster mitigation, ecological value, and tourism value, were integrated based on local potential and indigenous wisdom. This adaptation aligns with the premise that CES contributes significantly to socio-cultural sustainability [14], varies according to local socio-cultural contexts [16], and evolves by explicitly acknowledging the intricate relationships between humans, landscapes, and species [19]. The benefit categories and their operational definitions are detailed as follows.

2 Methodology

2.1 Study Area

The research was conducted on Weh Island, Sabang City. Geographically, Sabang is situated between the coordinates 95°13'2''–95°22'36'' East Longitude and 5°46'28''–5°54'28'' North Latitude. The administrative area of Sabang City encompasses five islands: Weh Island, Klah Island, Rondo Island, Rubiah Island, and Seulako Island, covering a total area of 122.33 km². The region is divided into two districts (kecamatan), eight sub-districts (mukim), and 18 villages (gampong). Among these five islands, only Weh Island is inhabited and serves as the primary hub for community and economic activities. Furthermore, Weh Island is the westernmost territory of the Republic of Indonesia. Sabang City, the regional capital, is home to a national forest conservation area designated as a Nature Tourism Park (Taman Wisata Alam). According to data from the Central Bureau of Statistics (BPS), the population of Sabang City in 2020 was 43,391 people [20].

2.2 Working Methods

This study aims to analyze land-cover changes and community perceptions of CES within the conservation areas of Weh Island. Data were collected through two primary approaches: satellite imagery digitization and respondent surveys via structured questionnaires. Spatial data processing was conducted using ArcGIS 10.1 software, which included image extraction, topographic and geometric corrections, and georeferencing. Meanwhile, respondent data were analyzed using statistical methods facilitated by SPSS software. The sample size was determined using the Slovin formula, resulting in 100 respondents selected through a purposive random sampling technique.

The first phase of the study involved classifying land-cover changes in the study area. Satellite imagery and other secondary data were processed using Geographic Information System (GIS) techniques to analyze land-cover dynamics for the year 2023. The accuracy of the resulting landscape classification maps was validated using the ERRMAT module in TerrSet 2020. Validation points were established through a stratified random sampling approach, in accordance with the protocols specified in Eqs. (1) and (2).

$$\text{Percentage correct} = \frac{\sum_{k=1}^q n_{kk}}{n} \times 100 \quad (1)$$

$$\text{Kappa coefficient} = \frac{n \sum_{k=1}^q n_{kk} - \sum_{k=1}^q n_{k+} n_{+k}}{n^2 - \sum_{k=1}^q n_{k+} n_{+k}} \quad (2)$$

Annotation:

n : Total number of pixels in the sample.

n_{kk} : Diagonal element of the k -th row and k -th column in the contingency matrix, representing correctly classified pixels of class k .

n_{k+} : Row sum of the k -th row in the contingency matrix (total pixels classified as class k).

n_{+k} : Column sum of the k -th column in the contingency matrix (total reference pixels belonging to class k).

q : Total number of land-cover classes.

The questionnaire distribution began with a preliminary briefing for all respondents regarding the concept of CES. This stage was crucial to ensure a shared understanding of the variables and to enhance the accuracy of the responses. Data were tabulated using Microsoft Excel and analyzed using a Likert scale to determine the percentage distribution of variables influencing each component of the CES. Furthermore, regression analysis was performed via SPSS to identify the dominant variables within the dataset.

Descriptive analysis was employed to characterize the respondent demographics and their perceptions of CES within the conservation forest areas. This approach allowed for a comprehensive qualitative and quantitative description of the findings, presented through both narrative synthesis and tabular data.

To investigate the relationship between the regional landscape and CES, Spearman's Rank Correlation Analysis was utilized. This non-parametric statistical method was chosen because it does not require a normal distribution of data and relies on ranking to determine the correlation coefficient. The strength of the relationship was categorized based on the following Correlation Coefficient ρ thresholds:

- 0.00–0.25: Very weak correlation
- 0.26–0.50: Moderate correlation
- 0.51–0.75: Strong correlation
- 0.76–0.99: Very strong correlation
- 1.00: Perfect correlation (absolute)

The decision-making criteria for hypothesis testing were based on a significance level of < 0.05 (indicating a significant correlation) and > 0.05 (indicating no significant correlation).

Cluster analysis was performed to identify groupings of ecosystem service values that exhibit similar features regarding the provision of various services. This analysis was conducted using SPSS software, employing both non-hierarchical (K-means) and hierarchical clustering methods. The latter generates a dendrogram for visual interpretation. The Euclidean distance metric was utilized to measure the dissimilarity between observations. The Euclidean distance between observation $(X_{1i}; X_{2i}; \dots; X_{ki})$ and $(X_{1j}; X_{2j}; \dots; X_{kj})$ is calculated as follows Eq. (3):

$$D(i, j) = \sqrt{(X_{1i} - X_{1j})^2 + (X_{2i} - X_{2j})^2 + \dots + (X_{ki} - X_{kj})^2} \quad (3)$$

The final stage of the study involved valuing cultural service perceptions in relation to specific landscape or land-use categories. This analysis began with a frequency-based scoring system, where scores were calculated by aggregating the occurrences of specific values attributed to each spatial land category.

Following the scoring process, Principal Component Analysis (PCA) was performed to identify and categorize clusters of values within each CES component [21]. Two primary criteria were utilized to determine the extraction of significant components:

- Eigenvalue Criterion: Only components with an eigenvalue > 1 (Kaiser's Criterion) were considered statistically significant and retained.
- Scree Plot Analysis: The scree plot was examined to identify the point of inflection. Components located before the curve levels off (where the slope "bends") represent the significant underlying dimensions of the data.

3 Results and Discussion

The following section presents the findings from the spatial analysis of satellite imagery and the empirical data gathered through field observations and respondent surveys regarding the perception of CES on Weh Island, Sabang.

3.1 Land Use and Land Cover Classification

Based on the 2023 LULC classification analysis, the total study area encompasses 11,095.45 hectares. The spatial distribution is dominated by green spaces, with forest areas covering 5,203 ha (46.89%) and agricultural land spanning 3,111 ha (28.04%). Collectively, these categories account for 74.93% of the total area. This high proportion of vegetative cover reflects a significant commitment to environmental conservation and the provision of public space, while also indicating a high potential for natural habitat preservation and mitigating environmental impacts.

In contrast, built-up areas account for 497.69 ha (4.49%) of the total area. While this percentage is relatively small, it represents significant urban development that could exert future pressure on green zones and water bodies. Water bodies cover a limited area of 157 ha (1.41%), highlighting the scarcity of available water resources and the critical need for sustainable management. Furthermore, open land (grasslands) was recorded at 2,071 ha (18.66%), which represents a potential frontier for future urban expansion. The detailed breakdown of these LULC categories is presented in Table 2.

The 2023 LULC pattern map of Sabang City, Aceh, illustrates various LULC categories, where the overall high proportion of green areas signifies the dominant land share of Weh Island/Sabang City. Nevertheless, a balance must be maintained between the development of built-up areas and the preservation of green zones and water bodies. Spatial planning must account for this sustainability, and enhanced protection for water bodies is paramount. Furthermore, despite the relatively small extent of open land, sustainable utilization strategies are required.

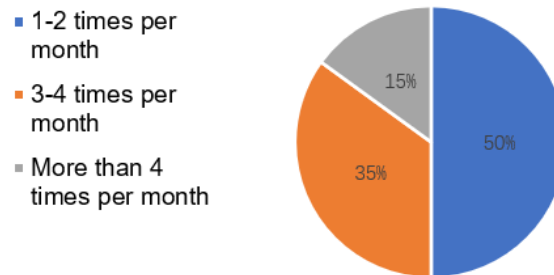
Table 2. LULC composition and spatial distribution on Weh Island

No.	Classifications	Year of 2023	
		Area (ha)	Percentage (%)
1	Built-up area	497.69	4.49
2	Woodland	5,203.00	46.89
3	Agricultural land	3,111.00	28.04
4	Grassland	2,071.00	18.67
5	Water body	157.00	1.41
6	Bare land	55.76	0.50
Total		11,095.45	100.00

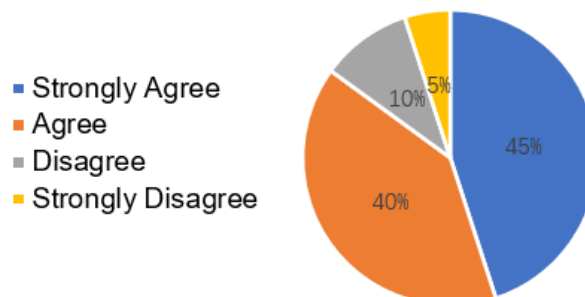
Note: LULC = land use and land cover.

3.2 Public Perception of Cultural Ecosystem Services

All respondents in this study reported having visited the conservation forest areas, engaging in a diverse range of activities. The presence of these conservation zones in Sabang significantly benefits the daily activities and well-being of the island's inhabitants. Regarding the frequency of interaction with the forest environment, the data indicate that 50% of respondents visit the conservation forest 1–2 times per month, 35% visit 3–4 times per month (Figure 3), and 15% visit more than four times per month.

**Figure 3.** Visitation frequency to the conservation forest

Regarding recreational value, respondents' feedback on the presence of the conservation forest area showed that 45% strongly agreed and 40% agreed with its use for recreational activities. In comparison, only 10% disagreed, and 5% strongly disagreed (Figure 4). It indicates that the conservation forest area effectively functions as a venue or medium for the community's recreational pursuits.

**Figure 4.** The role of conservation areas in supporting local recreational activities

Regarding the cultural heritage value associated with conservation areas, the survey results indicated that 25% of respondents strongly agreed and 50% agreed that these areas facilitate cultural activities. Conversely, 25% of respondents expressed disagreement. This distribution suggests that historical or cultural heritage significance is not uniformly perceived or present across all conservation forest zones in the study area (Figure 5).

Regarding the aesthetic value dimension, specifically the role of conservation forests in providing beauty through biodiversity and various local plant species, the responses obtained were as follows: 20% strongly agreed, 55%

agreed, and 25% disagreed (Figure 6). While significant portions of the conservation forests on Weh Island still possess rich biodiversity featuring indigenous flora, some community members remain unfamiliar with the specific local plant species found in the area.

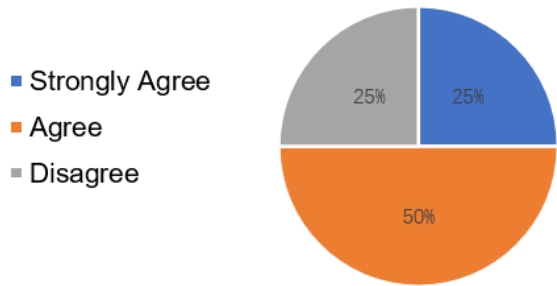


Figure 5. Public perceptions of cultural heritage values within conservation zones

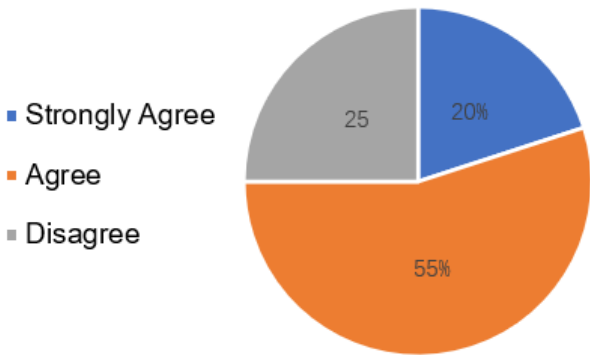


Figure 6. Graph of the aesthetic value provided by the presence of conservation forests

Regarding educational value, where the conservation forest serves as a venue for educational activities—specifically environmental research, the respondent feedback was as follows: 65% strongly agreed, and 35% agreed. The conservation forest area of Weh Island is frequently utilized for research activities (Figure 7).

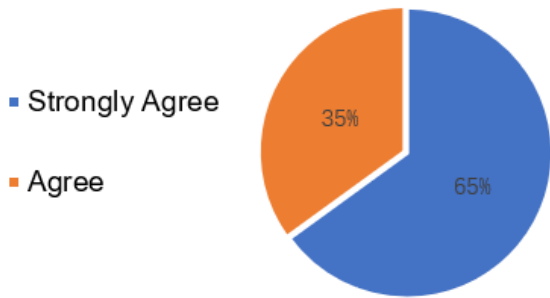


Figure 7. Conservation areas in educational value

The remaining CES dimensions, corresponding to the variables defined in Table 3, are summarized in the table below based on respondent perception levels.

Based on the data presented in Table 3, the perceptions of the Sabang community regarding CES within forest conservation areas demonstrate a high degree of agreement (Strongly Agree/Agree) across nearly all components: recreation, culture, aesthetics, education, health, spirituality, ecology, and tourism [22]. These results reflect a predominantly positive perception of the diverse benefits provided by CES in the conservation forests of Weh Island, Aceh, Indonesia. Its argument is further reinforced by the fact that local perceptions strongly support the integration of CES values into the “green and blue” planning of urban areas. This alignment suggests a growing public awareness of the intrinsic benefits of CES, recognizing that these services have become an integral part of their socio-cultural identity and long-term environmental resilience [23].

Table 3. Distribution of respondent perceptions across remaining CES categories

No.	Variables	Respondent Responses			
		Strongly Agree	Agree	Disagree	Strongly Disagree
1	Recreational value	0.45	0.40	0.10	0.05
2	Cultural heritage	0.25	0.50	0.25	0
3	Aesthetic value	0.20	0.55	0.25	0
4	Educational value	0.65	0.35	0	0
5	Social value	0.40	0.35	0.25	0
6	Health value	0.45	0.55	0	0
7	Spiritual value	0.45	0.55	0	0
8	DMV	0.10	0	0.20	0.70
9	Environmental/ecological value	0.40	0.55	0.05	0
10	Tourism value	0.60	0.35	0.05	0

Note: CES = Cultural Ecosystem Services; DMV = Disaster mitigation value.

In stark contrast to other categories, the Disaster Mitigation Value (DMV) received a predominantly negative response. Approximately 70% of respondents strongly disagreed, and 20% disagreed with the notion that conservation forests provide effective disaster mitigation, while only 10% strongly agreed. The community generally perceives these areas as inadequate for mitigation or evacuation purposes during a disaster. This negative perception is likely influenced by concerns regarding personal safety and the risk of criminal activity within the forest [24].

This argument is further substantiated by the Panglima Danau (the traditional authority of the Sabang lake region), who noted that illegal logging activities have transformed parts of the conservation forest into a source of environmental hazards, such as flooding and landslides, rather than a protective buffer. Furthermore, the strong religious convictions of the predominantly Muslim community in Sabang play a significant role; there is a profound belief that salvation and protection originate solely from God (Allah). This theological perspective reduces the community's reliance on, and perception of, the forest as a primary physical tool for disaster mitigation.

3.3 The Relationship Between Community Satisfaction Levels and Cultural Ecosystem Services

The relationship between the level of community satisfaction and the perceived values of CES was analyzed using Spearman's Rank Correlation. The results of the statistical analysis are as follows: The analysis yielded a significance value (p -value) of <0.001 . Since the significance level is less than 0.05, it is concluded that there is a statistically significant correlation between the Level of Community Satisfaction and Recreational Value. Furthermore, the Correlation Coefficient was found to be 0.491. Based on the predefined classification scale, this value indicates a moderate and positive (direct) relationship between the two variables (Table 4).

Table 4. Spearman's rank correlation results between community satisfaction and recreational value

Spearman's ρ		Satisfaction Level	Recreational Value
Satisfaction Level	Correlation Coefficient	1.000	0.491**
	Sig. (2-tailed)	-	<0.001
	N	100	100
Recreational Value	Correlation Coefficient	0.491**	1.000
	Sig. (2-tailed)	<0.001	-
	N	100	100

Note: "-" = not applicable; ** Correlation is significant at the 0.01 level (2-tailed).

There is a significant and relatively moderate correlation between the Community Satisfaction Level and Recreational Value, exhibiting a positive (direct) relationship. It suggests that as the perceived recreational value increases, the community satisfaction level rises accordingly [25].

Based on Table 5, only the DMV demonstrates no correlation with the level of community satisfaction. Conversely, all other variables correlate with the community satisfaction level. The variable with the highest correlation is the cultural heritage value, with a correlation coefficient of 0.591, which is categorized as a strong correlation.

This variable maintains a direct relationship with community satisfaction, signifying that an increase in cultural heritage value corresponds to a higher level of community satisfaction [26]. Meanwhile, the variable with the lowest correlation is the environmental/ecological value, with a correlation coefficient of 0.206, which is classified as a very weak correlation. Additionally, this variable has a direct relationship with community satisfaction, meaning that the higher the environmental/ecological value, the higher the community satisfaction level [27].

Table 5. Spearman's rank correlation matrix between community satisfaction and CES components

Variable 1	Variable 2	Correlation Coefficient	Sig.	Interpretation of Results		
				Correlation	Correlation Strength	Direction
Level of community satisfaction	Recreational value	0.491	<0.001	significant correlation exists	Moderate correlation	Positive (+)
	Cultural heritage value	0.591	<0.001	significant correlation exists	Strong	Positive (+)
	Aesthetic value	0.369	<0.001	significant correlation exists	Moderate correlation	Positive (+)
	Educational value	0.392	<0.001	significant correlation exists	Moderate correlation	Positive (+)
	Social relations value	0.547	<0.001	significant correlation exists	Strong	Positive (+)
	Health value	0.518	<0.001	significant correlation exists	Strong	Positive (+)
	Spiritual value	0.418	<0.001	significant correlation exists	Moderate correlation	Positive (+)
	DMV	-0.091	0.370		No correlation	
	Environmental/ecological Value	0.206	0.040	significant correlation exists	Very weak correlation	Positive (+)
	Tourism value	0.354	<0.001	significant correlation exists	Moderate correlation	Positive (+)

Note: CES = Cultural Ecosystem Services; DMV = Disaster mitigation value.

3.4 The Relationship Between Regional Landscapes and Cultural Ecosystem Services Across Land Use and Land Cover

The procedure for identifying the relationship between the regional landscape and CES through PCA is executed in the following systematic stages:

(a) Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy

The KMO test measures the proportion of variance in the variables that common factors can explain. In other words, the KMO helps determine whether the correlations among variables are strong enough to conduct factor analysis or PCA. KMO values range from 0 to 1. If the KMO value is high and the Bartlett's Test is significant (p -value < 0.05), then the dataset is suitable for PCA (Table 6).

Based on the output above, the KMO MSA value is $0.616 > 0.5$, and the Bartlett's Test of Sphericity (Sig.) showed $p < 0.001$; therefore, the analysis can proceed as it meets the required criteria.

(b) Communalities

Communalities in PCA measure the proportion of variance in each original variable that can be explained by the principal components generated from the analysis. In other words, communality indicates the extent to which the information or variability of a specific variable is represented within the PCA results.

Table 6. Kaiser-Meyer-Olkin (KMO) and Bartlett's test

Test	Statistic	Value
KMO Measure of Sampling Adequacy	–	0.616
Bartlett's Test of Sphericity	Approx. Chi-Square df Sig.	711.94 45 <0.001

Note: – = not applicable.

A communality value approaching 1.0 or exceeding 0.50 signifies that the retained components account for the majority of that variable's variance. Conversely, a communality value below 0.50 indicates that the variable is not well-explained by the selected components (Table 7).

Table 7. Communalities

Values	Initial	Extraction
Recreational value	1.000	0.680
Cultural heritage value	1.000	0.942
Aesthetic value	1.000	0.944
Educational value	1.000	0.805
Social value	1.000	0.928
Health value	1.000	0.686
Disaster mitigation value (DMV)	1.000	0.424
Spiritual value	1.000	0.784
Environmental value	1.000	0.982
Tourism value	1.000	0.926

Based on the statistical calculations, the initial communality for each variable is equal to 1.0, and the extraction communality for every variable exceeds 0.50. Consequently, these variables maintain a strong relationship with the selected principal components. High extraction communality values indicate that the variables are well-represented by the principal components and contribute significantly to the overall PCA. Only the disaster mitigation communality value of 0.424—which does not exceed 0.50—indicates weak communality value.

(c) Total Variance Explained

The total variance explained in PCA is a table that displays information regarding the amount of variance in the data explained by each principal component generated from the analysis. This table provides insights into how effectively the selected principal components summarize the information from the dataset. Based on the statistical output, the selection of components is determined by those with an initial eigenvalue greater than 1.0. The percentage of variance indicates the contribution of each component, while the Cumulative (%) represents the total amount of information captured by the retained components (Table 8).

Table 8. Total variance explained

Component	Initial Eigenvalues			Extraction Sums of Squared		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.661	56.611	56.611	5.661	56.611	56.611
2	2.440	24.399	81.009	2.440	24.399	81.009
3	0.965	9.652	90.661	–	–	–
4	0.754	7.540	98.201	–	–	–
5	0.149	1.491	99.692	–	–	–
6	0.031	0.308	100.000	–	–	–
7	2.967E-16	2.97E-12	100.000	–	–	–
8	9.04E-14	9.04E-13	100.000	–	–	–
9	-4.89E-14	-4.89E-13	100.000	–	–	–
10	-3.53E-13	-3.53E-12	100.000	–	–	–

Note: – = not applicable.

(d) Component Matrix

The Component Matrix in PCA displays the factor loadings for each variable across the principal components generated by the analysis. These factor loadings indicate the strength of the relationship or correlation between the original variables and each principal component. Components are typically selected based on established criteria, such as an eigenvalue > 1 or the visual interpretation of the scree plot. The Component Matrix illustrates the correlation between the original variables and the derived principal components (e.g., Component 1 and Component 2) explicitly. In this matrix, each column represents a single component, while each row corresponds to an individual variable (Table 9).

Table 9. Component matrix

Values	Component	
	PC1	PC2
Tourism value	0.915	0.298
Educational value	0.897	-0.017
Health value	0.807	-0.183
Cultural heritage value	0.804	-0.544
Aesthetic value	0.767	0.597
Social relations value	0.747	-0.609
Environmental/ecological value	0.745	0.654
Spiritual value	0.665	0.584
Recreational value	0.593	-0.573
Disaster mitigation value (DMV)	-0.476	0.444

Extraction Method—PCA

(1) The PCA resulted in the extraction of two primary dimensions, designated as PC1 and PC2. The factor loadings within the matrix are interpreted as follows: High Positive Loadings (+):

(2) These indicate a strong direct correlation between the variable and the component. A higher positive value suggests that the variable is a primary driver of that specific component's identity. High Negative Loadings (-): These signify a strong inverse relationship. While the variable is still a significant contributor to the component, its influence moves in the opposite direction compared to positive variables. Values Approaching Zero: These indicate that the variable does not provide a significant contribution to that particular component and does not represent a defining characteristic of that dimension.

The plot type is a biplot, which combines the representation of variables (indicated by red dots) and observations or locations (indicated by blue dots) in a single graph. Bar Chart Representation Tool (BART) is a bar chart that explains the position of objects on the x-axis and y-axis which functions to visually compare categories. Based on the generated scatter plot (Figure 8), the results, interpretations, and implications regarding the correlations between CES and conservation forests are presented as follows:

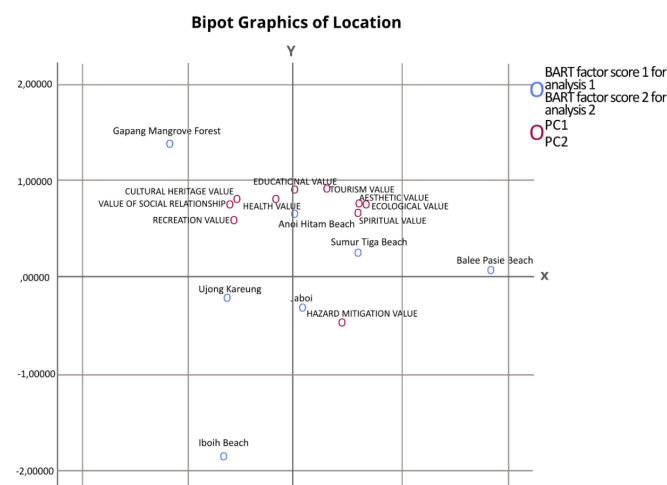


Figure 8. Scatterplot of the relationship between Cultural Ecosystem Services (CES) and the landscape on Weh Island

Results

Locations are scattered throughout the plot, indicating variation in their characteristics. Several conservation area locations, such as the Gapang Mangrove forest and Iboih Beach, are situated far from the center, signifying that they possess significantly different characteristics. Variables that are adjacent or form a slight angle are positively correlated. For instance, Aesthetic and Environmental Spiritual values appear to be positively correlated. Variables pointing in opposite directions (forming a wide angle) tend to be negatively correlated; an example is the relationship between Educational and DMV. Variables that form a right angle are uncorrelated with each other.

Locations situated near the direction of a variable tend to have high values for that specific variable. For example, the conservation forest area at Balee Pasie Beach appears to have high values for BART factor scores 1 and 2. Conversely, locations positioned in the opposite direction of a variable tend to have low values for that variable. The associations between locations and variables, as well as the correlations between variables for each location based on the displayed biplot, show that the Gapang Mangrove Forest location is situated far in the upper-left of the plot, strongly positively correlated with Cultural Heritage and Social Relation values, while negatively correlated with Aesthetic, Environmental Spiritual, and DMV. Iboih Beach is located in the lower-left portion of the plot, correlating positively with recreational values, and strongly negatively correlated with Educational and Tourism values.

The Ujong Kareung Beach conservation area is located in the lower-left-center of the plot, tending to be neutral toward most variables and slightly positively correlated with Recreational values. The Jaboi area is situated near the center of the plot, slightly to the lower right. It tends toward neutrality regarding most variables, exhibiting a slight positive correlation with DMV. Furthermore, Anoi Hitam Beach and Sumur Tiga Beach are both situated close to each other in the center-right of the plot, indicating a positive correlation with Aesthetic, Spiritual, and Environmental values, while being negatively correlated with Cultural Heritage and Social Relation values. Balee Pasie Beach, which is located far to the right of the plot, is strongly positively correlated with BART factor scores 1 and 2 and negatively correlated with Cultural Heritage and Social Relation values.

Interpretation

Based on the displayed scatter plot, an analysis can be conducted regarding the landscape categories in relation to CES. CES encompass the non-material benefits that humans derive from ecosystems, including aesthetic, spiritual, educational, health, recreational, and cultural heritage values. The following is an interpretation of the relationship between landscape categories and CES:

- The Gapang mangrove forest conservation area has a strong association with CES related to cultural heritage and Social Relations values, while being relatively weak in terms of Aesthetic and Environmental Spiritual aspects. It indicates that the mangrove forest possesses high value as cultural heritage and facilitates social relationships through traditional practices or communal values. Local community perception suggests that the forest area, particularly the mangroves, is viewed as a site rich in cultural values that facilitates social interaction [28].

- Coastal conservation areas (Iboih Beach, Anoi Hitam Beach, Sumur Tiga Beach, and Balee Pasie Beach) demonstrate significant variation in CES. For instance, Iboih Beach is a natural beach where ecosystem services are dominant in the Recreational aspect. Local community perceptions identify this area as a site for relaxation and recreational activities, with potential for the development of eco-friendly recreational facilities [26].

- The Anoi Hitam Beach and Sumur Tiga Beach conservation areas exhibit stronger correlations with CES related to Aesthetics, Health, and Environmental Spirituality. The Health component is closely linked to socio-cultural aspects, including cultural heritage, social interaction, and recreation. It suggests that, within this context, health aspects are influenced by or interact with cultural activities, heritage, and recreation at these specific sites. Local community perceptions regard the area as a site of aesthetic beauty and spiritual significance, offering potential for development that preserves natural integrity while facilitating spiritual practices or meditation [29].

- The Balee Pasie Beach conservation area is highly developed, as indicated by its high BART factors, signifying a high level of modernization. It indicates that the beach can provide various CES, ranging from recreation to spiritual values, depending on its specific characteristics and level of development. The local community perceives this area as more advanced or modern, with development potential centered on tourism infrastructure, while maintaining a focus on sustainability [30].

- The Jaboi Volcanic Area conservation site is located near the center of the plot, showing a relative balance across various CES. It is slightly stronger in the Disaster Mitigation aspect, which is logical for a volcanic area. The local community perceives this area as one that requires special attention regarding safety, yet also possesses significant tourism potential. There is potential for developing educational programs on volcanology and geological tourism, while also considering safety aspects [31].

- Urban/Residential Areas (not explicitly shown, but likely associated with BART factors); locations with high BART factor scores (such as Balee Pasie Beach) may represent more developed or urbanized areas. These areas are weak in terms of Cultural Heritage and Social Relations, but may be stronger in Education and Tourism. Local community perceptions view these areas as centers for economic and educational activity [32].

Planning Implication

The recommended planning implications are as follows:

- Conservation forest landscapes, which provide diverse CES, must have their biodiversity preserved and inventoried. Furthermore, these ecological assets must be integrated as key elements within regional spatial planning documents.
- As Sabang City focuses on developing tourism, it is imperative to establish conservation forests as key destinations to foster and realize the welfare of the local community.

4 Conclusion

Based on the analysis above, the relationship between landscape categories and CES in Pulau Weh, as shaped by community perception, can be concluded as follows:

- More natural or traditional landscapes (such as mangrove forests) tend to be strong in cultural heritage and social relations, while more developed landscapes tend to be weak in these aspects but may be stronger in education and tourism.
- Based on the plot, CES are closely linked to various landscape aspects, including historical sites, recreational locations, and scenic natural landscapes. The interconnection between cultural variables across different locations underscores the importance of conserving natural and cultural landscapes, as they provide space for social interaction, spiritual activities, and aesthetic appreciation, all of which are vital for human well-being.
- Some coastal forests exhibit high aesthetic and spiritual values, while others (such as mangrove forests) are relatively weak in these aspects. It indicates that aesthetic and spiritual values do not consistently correlate with the degree of landscape naturalness.
- The health component is viewed within a holistic landscape, where the environment, cultural heritage, and social interaction all provide significant contributions. When associated with specific geographical landscapes, sites that possess both recreational and cultural heritage aspects also tend to integrate health considerations.
- Coastal forest landscapes exhibit significant variation in recreational dimensions; some areas, such as Iboih Beach, focus primarily on natural recreation, while others, such as Balee Pasie Beach, are characterized by higher levels of infrastructural development.
- The disaster mitigation aspect emerges as a relatively independent service, exhibiting limited correlation with other CES. However, it appears more pertinent to specific landscape types, such as volcanic areas.
- This analysis facilitates landscape planning and management to maximize various CES while maintaining a balance between conservation, development, and local cultural values.

Author Contributions

Conceptualization, Z.F. and A.A.; methodology, Z.F.; software, Z.F.; validation, Z.F., A.A., E.W., and M.I.; formal analysis, Z.F.; investigation, Z.F.; resources, Z.F. and A.A.; data curation, Z.F.; writing—original draft preparation, Z.F.; writing—review and editing, Z.F. and A.A.; visualization, Z.F.; supervision, A.A. and E.W.; project administration, Z.F. and M.I.; funding acquisition, Z.F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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